Early warning for problematic gaming

Capstone literature, data and technology review

Project name Problematic Gaming Early Warning
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This project investigates whether gaming behaviors observed early on contribute to predicting later problematic gaming symptoms. The core question is whether detailed telemetry provides a measurable predictive contribution beyond the baseline Gaming Disorder Test (GDT) score and total playtime. The primary target is the change between two GDT measurements; the secondary target is the later measurement of how much gaming interferes with daily life.

The report preserves the idea of the shared project PDF and presentation and makes the research design concrete with Eda Nur Saruhan's feedback. The literature and source-code review is complete, and preliminary statistics have been computed from the survey files that were accessed. Telemetry features have not yet been generated, and no model training or performance evaluation has been carried out. The methods and product sections below describe the planned work.

1 Literature review

1.1 Problem and target population

The problem examined is the change in symptoms associated with gaming becoming harder to control and negatively affecting daily functioning. The WHO's ICD-11 framework emphasizes loss of control, giving gaming priority over other activities, and continuing despite negative consequences. Diagnosis requires marked impairment in functioning; playing for long hours alone is not sufficient (World Health Organization, 2020). The project will investigate the symptoms on the scale rather than clinical diagnosis.

The primary analysis group consists of US and UK players aged 18–40 whose eligibility for participation in Open Play has been verified. Although university students are a possible user group, the data do not consist only of students. No direct generalization will be made to students in Türkiye, to children, or to people seeking clinical help. Signals of disruption related to work or school, sleep patterns, and social life make the research important.

1.2 Structure and purpose of the review

The focused narrative review is organized around three themes: symptom measurement, the contribution of objective gameplay logs, and forward-looking prediction. Peer-reviewed foundational studies, the WHO source, and Open Play's data documentation were examined together. This work is not a comprehensive systematic review. The literature is needed to choose an appropriate target, to avoid confusing high engagement with disorder, and to avoid presenting a concurrently measured relationship as a prediction of the future.

1.3 Summary of key studies

Development of GDT measurement

Pontes et al. (2021; first published online 2019) developed the four-item GDT with 236 Chinese and 324 British gamers. Online surveys, factor analysis, and validity and reliability assessments were used. The findings supported a single-factor structure. The study provides a measurement basis for our project's outcome variable; however, validity in an online community sample does not by itself imply an individual clinical diagnosis. The three-month recall period in Open Play should also be treated as an interpretive limitation specific to this project.

The gap between self-report and usage logs

Parry et al. (2021) combined 106 effect sizes from different digital media studies in a preregistered meta-analysis. Here the sample is not a single group of players; the unit of analysis is the effect sizes obtained from prior studies. Only moderate agreement was found between self-reported and logged use. The result supports deriving playtime from telemetry whenever possible. Because different media and measurement methods were pooled, the findings do not directly show how well GDT change can be predicted.

Playtime and problematic gaming symptoms

Jin et al. (2022) matched the surveys of 320 North American League of Legends players with actual gameplay logs. The mean age was 21.84 and the proportion of men was 79.1%. The correlation between self-reported and logged weekly time was 0.494, and the mean absolute difference was 5.71 hours. Both time measures were associated with symptoms, with small but significant relationships. The study makes a comparison model based on total time necessary; because it covers a single game and has a cross-sectional design, it provides no evidence on later symptom change.

Objective use and well-being

Vuorre et al. (2022) linked six weeks of objective gameplay data from 38,935 players to three well-being measures. In longitudinal data from seven games, little evidence was found for an effect of average playtime on well-being. This result supports not treating time alone as an indicator of harm. However, the study's target was general well-being, not the GDT or clinical gaming disorder. The finding therefore cannot be interpreted as meaning that no problem exists in high-risk subgroups.

Forward-looking risk prediction with machine learning

Stavropoulos et al. (2023), in a two-wave study with a baseline sample of 565 players and a six-month interval, used the player–avatar bond, age, and gaming experience. Tuned and untuned classifiers were compared; Random Forest gave the best results, and immersion in the avatar was an important predictor. The study shows that future GDT risk can be investigated. However, its main inputs are self-reports; it does not directly test the additional contribution of detailed multi-platform telemetry beyond the baseline GDT. We will not assume that the same model will perform best in our sample.

1.4 Synthesis and research gap

Direct link to Open Play

Ballou et al. (n.d.) present Open Play, which brings together multi-platform gameplay logs and psychological measures. The data documentation covers approximately 2,000 survey participants; the larger group with verified eligibility comprises 2,238 people. This source was treated as a data introduction and preprint, and is not presented as a peer-reviewed study of predictive performance. The authors state as an open research opportunity whether irregular, long, or late-night gaming predicts later increases in problematic gaming. This directly justifies our project's question; it does not confirm the outcome in advance.

Comparison of the studies

Jin et al. and Parry et al. agree on the limitations of measuring the amount of use by self-report. While Jin et al. found a small association between playtime and symptoms, Vuorre et al. report very small average effects for general well-being. Because the target variables, samples, and designs differ, these are not directly contradictory results. Well-being, playtime, and gaming disorder symptoms should not be used interchangeably.

Pontes et al. show how symptoms can be measured, and Stavropoulos et al. show how such measurements can be turned into a forward-looking prediction task. The shared limitations are voluntary participation, samples selected by community or by game, self-report-based outcome measurement, and observational design. None of these studies proves, for our own data, the additional contribution of the early telemetry window we will select beyond the baseline GDT.

The gap the project addresses

Our contribution is to conduct an open and reproducible added-value test using multi-platform behavioral features. Besides total time, gameplay timing, irregularity, and time trend will be examined. Feature groups will be compared on the same people, at the same prediction time, and on the same evaluation splits. We do not claim to be the first early warning system in the field; we aim to test a specific research gap with accessible data.

Research questions

RQ1 Does gaming behavior in the first 14 days, taking the baseline GDT and total playtime over the same period into account, better predict the GDT measurement about 10 weeks later and the change between the two measurements?

RQ2 Can the 14 days of behavior before a survey predict the interference with daily routines reported in the next survey, beyond the current level of interference and total time?

RQ3 Does classifying the open-ended responses given at baseline provide information in addition to the numerical variables? This question is a secondary extension that depends on text coverage.

Conclusion of the literature section

The most defensible design is to test regularized tabular models against strong comparisons. Success is not merely a high prediction score but a consistent added contribution from telemetry. If no such contribution is found, the result is still valuable: it may show that short scales can be sufficient instead of complex monitoring. Developing an effective intervention would require separate user and effectiveness research after the prediction study.

2 Data research

2.1 Source selection and scope of review

Open Play was chosen because it combines behavioral logs with psychological measures from the same individuals. The co-availability of the GDT, effects on functioning, and open-ended texts suits our research questions. The clean CSV files, codebook, and cleaning script in the repository were reviewed. In this preliminary review, the biweekly, intake, iOS, and Android files were downloaded. Feature extraction from console and Steam telemetry is the next stage (Digital Wellbeing, n.d.-a, n.d.-b, n.d.-c).

The calculations are pinned to the GitHub version below, accessed on 11 October 2026. Because later changes to the main branch could affect the results, the same version must be used for reproduction. The Zenodo DOI is provided to reach archived versions; the numbers in this report were not computed from a Zenodo archive.

GitHub commit eb0a68c1a7ca1fad54e57a1f89b9d5450f8be840

File	Review finding	Role in the project
survey_biweekly.csv.gz	6,844 rows, 158 columns, 1,948 people	GDT, effects on life domains, open-ended responses
survey_intake.csv.gz	33,046 rows, 88 columns	Screening pool; eligibility, age, country, and time zone
ios.csv.gz and android.csv.gz	3,608 and 2,518 rows; 4 columns each	Mobile daily time; contains no sessions or night-time hours
codebook.xlsx and data-process.qmd	Variable dictionary and transformation code	Verifying coding, timing, and cleaning decisions

2.2 Time span and collection method

The survey design includes six waves every two weeks within a 12-week schedule; the nominal gap between waves 1 and 6 is 10 weeks. The calendar range of the accessed biweekly file is 1 October 2024 to 27 September 2025; this does not mean each person was followed for a year. Among the 710 selected people, the mean interval between GDT measurements is 71.22 days. Older telemetry history varies by platform; it should not be confused with the 12-week survey follow-up.

Platform	How the record is obtained and its resolution	Eligible features
Nintendo Switch	Platform data sharing; sessions of games published by Nintendo	Daily time, session length, and local time
Xbox	Platform sharing with participant consent; sessions	Time, session, and hour; anonymized game ID, genres
Steam	Hourly changes in total time via API	Daily and hourly patterns; sessions approximate
iOS and Android	Daily totals via OCR from screen-time screenshots	Daily time and day-to-day change

Source: Ballou et al. (n.d.) and Digital Wellbeing (n.d.-b). Platforms do not offer the same level of detail; it will not be assumed that the same features are available on all of them.

2.3 Directly computed sample and GDT findings

The proposed cohort consists of people aged 18–40 with qualified=True in the intake file. The planned six waves were used. The four GDT responses were coded as Never=1, Rarely=2, Sometimes=3, Often=4, and Very often=5; measurements with all four items completed were summed. The total score ranges from 4 to 20. Missing targets were not imputed.

Selection step	Number of people
With at least one record in the biweekly file	1,948
With all four GDT items completed in waves 1 and 6	730
Of these 730 people, with a matching intake ID	723
Additionally with qualified=True	710
Additionally meeting the 18–40 age requirement	710

These 710 people are a survey cohort to which no telemetry observation-coverage condition has yet been applied. The final model sample may shrink to those with sufficient telemetry in the selected early window.

Measure	Wave 1	Wave 6	Change
Mean ± standard deviation	8.44 ± 3.25	7.63 ± 3.12	−0.81 ± 2.70
Median	8	7	−1
Quartiles and range of change	—	—	Quartiles −2 and 1; range −12 to 10

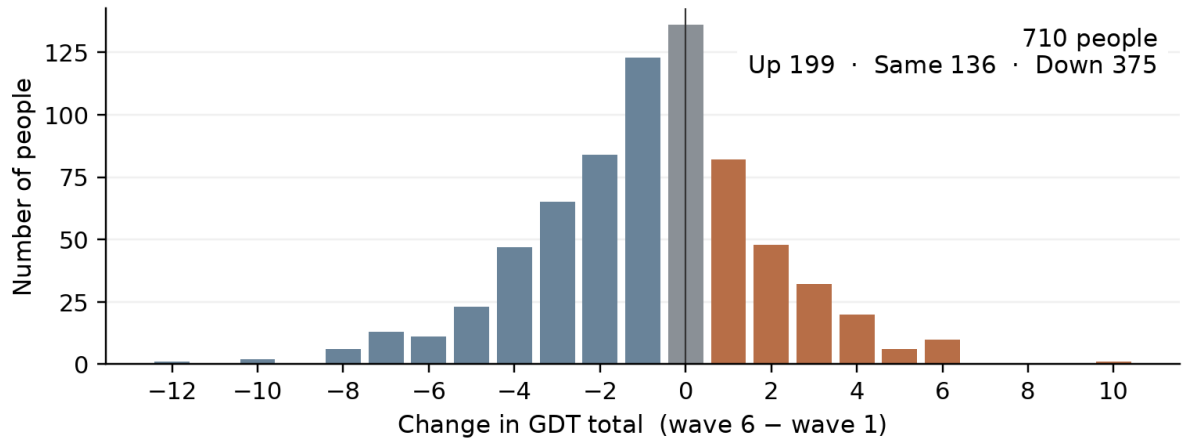


Figure 1 GDT total difference for the same 710 people. Score increased for 199 people (28.0%), unchanged for 136 (19.2%), decreased for 375 (52.8%). Source: computed for this report from the pinned Open Play files.

2.4 Secondary target and data-quality findings

Interpreting the GDT findings

Of the 1,663 baseline participants in the cohort, 1,598 have a valid baseline GDT; 710 of them provide both measurements (44.4%). The baseline mean of those who completed both measurements is 8.44, while the mean of the 888 people without a final GDT is 8.88. This descriptive difference means attrition should not be assumed to be random. The mean decrease in score is not the effect of an intervention. An increase alone is also not considered clinically meaningful deterioration. The correlation between the two GDT scores is 0.642; this value is not the model's test performance.

Wave	All survey rows	Selected cohort rows	Valid GDT
1	1,948	1,663	1,598
2	1,190	1,138	2*
3	1,111	1,073	0
4	964	933	0
5	880	856	0
6	740	720	710

* The two completed GDT records in the second wave will not be used as a planned new GDT measurement. In the whole file there are also 11 rows with wave=7. The relationship between the wave code and the time of administration will be traced using the cleaning code and the raw records.

Subjective displacement variables

These are found in the fields for effects on life domains: gaming_value_work, gaming_value_social, gaming_value_cognitive, gaming_value_emotion, and gaming_value_routines. In the codebook, responses are described as 1–7, whereas in the clean file they range from −3 to +3. Negative values indicate interference, positive values indicate support. Skipping this difference and automatically treating a high score as more harm reverses the direction of the target.

The priority secondary target is interference with daily routines: the sign of gaming_value_routines will be reversed so that a higher value indicates more interference. The item covers sleep, hygiene, and eating routines together; it is not only a sleep measure. Impact on work or school will be a separate exploratory outcome. The sum of the five domains will not be presented as a validated clinical scale.

In the selected cohort, there are 6,295 surveys with all five of these domains completed. From the 1,206 people with responses in two consecutive waves, 4,372 person–transition pairs are obtained. When a minimum 14-day gap between measurements is imposed to prevent the two-week recall period of the later survey from extending before the prediction time, 2,775 pairs remain from 1,161 people. These numbers are also before the telemetry coverage filter; 2,775 does not mean independent individuals.

2.5 Prediction time and behavioral features

Time boundary of the primary target

In the primary analysis, the prediction time will be fixed at 14 days after the person's wave 1 survey. Only telemetry from this 14-day early period and baseline variables known at the prediction time will be used. GDT change is the main output of the study. The model will use the baseline GDT as an explanatory variable to predict the wave 6 total; the change prediction will be produced by subtracting the baseline score from the predicted total. Predicting the difference directly will be a separate sensitivity analysis.

Important measurement limitation: In the codebook, the GDT questions ask about the last three months. The recall periods of the two measurements, about 10 weeks apart, overlap; the final GDT also covers the early telemetry period. The task is therefore to predict symptom burden measured later. We cannot claim that we detect only symptoms that newly begin after the prediction time or that we predict disorder onset. For a true forward-looking window test, the displacement analysis is more suitable.

Time boundary of the secondary target

When wave k is completed, the telemetry of the previous 14 days and the current interference score will be used to predict interference with routines in wave k+1. Telemetry at the time of the later survey or text from the same survey will not be added to the input. Actual date ranges will be used; the minimum 14-day rule and different window lengths will be reported in the sensitivity analysis.

Feature group	Planned computation	Scope and interpretation
Total time and frequency	Hours scaled to 7 days; active-day ratio; consecutive active days	Based on observed days; core comparison
Daily irregularity	Standard deviation and interquartile range of daily time	Depends on daily coverage across all platforms
Night-time gaming	Share of gaming minutes between 00:00 and 06:00 local time	Steam, Nintendo, Xbox; alternative hour definitions will be tried
Long sessions	Session median, upper percentile, and share of sessions exceeding 2 hours	Nintendo and Xbox; the threshold is not a clinical limit
Weekly pattern and trend	Weekend–weekday difference in daily time; second week minus first week	Work schedule and number of observations are taken into account
Game variety and genre	Number of unique games; time share of matched genres	Metadata matching and platform coverage are reported

Steam's hourly records will not be accepted as exact session starts and ends, and a night-time share will not be derived from mobile daily totals. Long-session findings will be interpreted bearing in mind that the existing cleaning rules may have removed extremely long records.

2.6 Missing data, ethics, and conclusion of the data research

Handling missingness and measurement differences

Absence of a record will be distinguished from genuinely not playing. An invisible Steam profile or a missing mobile screenshot will not be coded as zero playtime. Calendar and platform coverage will be built, coverage ratios will be recorded as features, and the minimum coverage condition will be set without looking at the results. For example, 80% could be considered as the minimum coverage, and 60% and 90% for the sensitivity check; these are not filters that have been applied yet.

Even after cleaning, the survey–intake match requires a separate check: 108 IDs in the biweekly file have no intake counterpart; seven of them completed both GDT measurements. The main cohort does not include them. The exclusion flow will be reported, and version compatibility will be examined rather than automatically treating those without eligibility information as eligible. Missing GDT labels will not be generated; any imputation needed in explanatory variables will be learned only from the training data.

Different platforms and games left open at the same time may affect total time. Hourly totals exceeding 60 minutes, sessions crossing the day boundary, daylight-saving transitions, and outlier durations will be audited. For local time conversion, the repository's approach will be followed; it will be kept in mind that using only a fixed GMT offset can cause errors during daylight saving time. The contribution of detailed features will be compared with the baseline model on the same subsample.

Ethical privacy and access conditions

The data are shared under CC0 terms with an additional no-reidentification clause. Open access does not mean that the data are fully anonymous or freely re-linkable. No attempt will be made to identify individuals; identities will not be linked with gaming accounts or external profiles (Digital Wellbeing, n.d.-d).

Only the necessary columns will be kept in the analysis; political views, detailed geography, and unnecessary health fields will not be brought into the working table. Access to the raw data will be limited to the project team, and files will be kept in an access-controlled environment. In presentations, person IDs, raw daily schedules, and recognizable open-ended texts will not be shared; aggregated results or clearly synthetic examples will be used. This approach is consistent with the principle of data minimization (Personal Data Protection Authority, n.d.).

No new participant data will be collected at this stage. If a user study is planned in Türkiye in the future, the purpose, data types, retention period, access, withdrawal, and sharing conditions will be explained; the necessary ethical review and an appropriate legal basis will be determined separately. Information notice and, where required, explicit consent will be handled separately. Children are outside the current scope.

Conclusion of the data research

A usable survey foundation exists for the main study, but planning should be done with a matched and eligible GDT cohort of 710 people rather than about 2,000. The more frequently measured functioning effects provide additional analysis. The next priority is to verify early telemetry coverage and finalize the model sample. No relationship or predictive contribution between telemetry and symptom change has yet been computed.

3 Technology review

3.1 Rationale for technology selection

A large number of data rows does not enlarge the number of independent people. For the main result, the structure is at most 710 people and is heterogeneous in terms of platform and missingness. The priority is therefore a limited number of interpretable features, regularization, and person-level evaluation. Technology selection will be made on the basis of explainability, development cost, data privacy, and reproducibility.

Approach	Strength	Limitation and decision in the project
Ridge and Elastic Net	Low computational cost; regularization; inspectable coefficients	Captures nonlinear relationships only to a limited extent. Primary baseline models.
Random Forest	Nonlinear relationships and interactions; little preprocessing needed	Tuning and depth matter in a small sample. Secondary comparison.
Gradient Boosting	Flexible relationship modeling on tabular data; controlled complexity	Risk of overfitting; narrow tuning range and internal validation. Main nonlinear candidate.
LSTM or Transformer	Potential to process dense sequential data directly	Few independent labels, more computation and tuning burden. Not selected in the first version.
SHAP and permutation importance	Explaining feature contribution to model output	Does not show causality; attribution may change among correlated features. Explanation of the final model.

This comparison is a design assessment, not a performance result. scikit-learn (n.d.-a, n.d.-c) was used as the basis for model properties and Lundberg and Lee (2017) for the explanation approach. It is not claimed that deep models fail under all conditions.

3.2 Implementation tools and cost

Reading compressed CSVs, date handling, and feature generation will be done with Python, pandas, and NumPy. Matplotlib will be used for descriptive plots, and the scikit-learn Pipeline for transformation, regularization, and validation. The work will be kept as a notebook or script that can be rerun with pinned data and dependencies. The current preliminary review was carried out in Python.

These tools can be run locally with open-source software; a paid model API is not required. Computation, storage, or optional server costs may arise separately; no monetary estimate is given until the budget and usage volume are determined. The first tabular models can be tried on a CPU. If the data grow, memory and processing load will be reduced by generating daily per-person features in advance.

Streamlit is a candidate for a small prototype that displays research results (Streamlit, n.d.). The first version will be a research interface that explains validated model output and past behavior, not a mobile application collecting real-time data. Live platform integrations would require separate permissions, API access, and maintenance work.

3.3 Model comparison and evaluation design

Measuring the added contribution of telemetry

Model	Information used	Purpose of the comparison
M0	Treat the wave 6 GDT as equal to the baseline GDT	Baseline control against the no-change assumption
M1	Baseline GDT only	Strong comparison learned from the baseline score
M2	M1 and the early period's total hours scaled to a week	Added contribution of simple gaming amount
M3	M2 and predefined behavioral features	Added contribution of timing and patterns
M4	M3 and labels extracted from the baseline open-ended text	Contribution of NLP; the same subsample that has text

The same eligible models, the same folds, and a similar tuning budget will be used for M1–M4. If detailed telemetry exists only on certain platforms, M2 will also be re-evaluated on that subsample. In this way, the effect of different people or more tuning trials will not be confused with the feature contribution. For the secondary target, the comparison is the model that carries the current routine-interference score forward to the next measurement, and its version with total time added.

Validation and prevention of data leakage

People will be split into training and evaluation sets; different waves of the same person will not fall on both sides. The initial plan is person-level cross-validation with five outer and three inner folds. The outer folds determine performance and the inner folds determine model tuning. For the secondary target, GroupKFold-like person groups are mandatory. Imputation, scaling, feature selection, and text vectorization will be learned on the training part of each fold (scikit-learn, n.d.-b).

All inputs must be available at the defined prediction time. The wave 6 GDT, the final-wave text, and telemetry after the prediction time will not leak into any early prediction input. In addition to generalization to new people, if enough data remain, training on earlier calendar cohorts and evaluating on later cohorts will be carried out. Calendar time and the person-specific wave number will be kept separate.

Success metrics and explainability

The primary metric will be mean absolute error (MAE), with RMSE and R^2 as auxiliary metrics. The person-level error difference between M3 and M2 and an uncertainty interval from resampling over people will be reported. When the same baseline score is subtracted, the absolute errors of the final GDT prediction and the change prediction are identical; R^2, however, depends on the variance of the target and must be computed separately. A small and uncertain contribution will not be presented as a strong success.

If an increase/no-increase classification is performed only as an exploratory step, it will be stated that a one-point increase is not a clinical threshold. For imbalance, PR-AUC, sensitivity, precision, the confusion matrix, and calibration will be examined; accuracy alone

will not be used. SHAP explanations will be produced for people unseen in the outer folds and will be interpreted together with correlated features and platform differences. A feature's contribution to the prediction does not prove that changing that behavior would reduce symptoms.

3.4 NLP use cases and implementation limits

Additional signal from open-ended responses

The problematic_play field in the codebook is an optional question asking for an explanation of gaming's negative contribution to life. Although the paper mentions monthly open-ended texts, in the downloaded clean file the responses are mainly in waves 1 and 6. In the selected cohort, these waves have 1,568 and 692 completed responses respectively; there are also two records in wave 2. It will therefore not be assumed that there are three separate monthly text measurements or six text waves.

For the NLP extension, baseline texts can be coded with multiple labels such as no negative impact, difficulty controlling, work or school impact, routine or sleep impact, social impact, and unclear response. First, a small sample will be labeled independently by two human raters, disagreements will be resolved, and a labeling guide will be prepared. An empty response will not be counted as a no negative impact label.

The simple comparison will be TF-IDF and regularized logistic regression. A language model will be added only as a candidate evaluated against this comparison with human labels. Macro-F1, label-level precision and recall, and the share of unclear responses will be reported. The language model's label will not be treated as a validated ground truth; the model version and prompt will be pinned. Raw texts will not be sent to an external API by default, and personal expressions will be removed (scikit-learn, n.d.-d).

Similar tools and real-world examples

Android Digital Wellbeing offers usage visibility and app timers; Apple Screen Time also provides time reports and usage limits (Google, n.d.; Apple, n.d.). These features are examples of behavioral awareness. The support documents reviewed do not show that these tools are validated systems predicting GDT change. Our project aims to add an explainable prediction dimension to them at the research level.

Stavropoulos et al. (2023) are a research example of future risk prediction with machine learning, and Open Play is a research example of a multi-platform data infrastructure. However, it will not be claimed that a classroom prototype is validated for clinical use or reduces addiction. For users seeking support in Türkiye, referral can be offered to YEDAM's free psychological and social support for technology addiction and to the 115 line (Yeşilay Counseling Center, n.d.).

Safe and accessible use in the prototype

Alongside total time, the interface will show which behaviors have changed and the model's uncertainty in short explanations in Turkish. It will not give predictions when data are insufficient; users will be able to turn off notifications and delete their data. Daily streaks, point rewards, or constant alerts will not be designed. An example explanation, only if the actual computation supports it, would take the form "An increase in the night-time gaming share contributes to a higher symptom prediction in this model"; it will not give a diagnosis or treatment recommendation.

Conclusion of the technology section

Regularized regression, gradient boosting, person-level validation, and SHAP are a suitable set of tools for the first version. NLP is an optional extension. The project's next concrete

output is an analysis table with verified telemetry coverage and the M0–M3 comparison. The model's usefulness can be discussed only after this evaluation.

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In-text citations and the reference list follow APA 7 author–date style. The access date for web sources that have no date and may change is 11 October 2026. The version links given for the Open Play files are for reproducing this report's numerical results.

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Archive access: Zenodo DOI for the Open Play data and code versions. The calculations in this report belong to the GitHub version pinned above.

https://doi.org/10.5281/zenodo.17536656